EXERCISE 6

Exactness of Separated Differential Forms

Original statement · ProblemsCh2.pdf, p. 2

Exact · Separated form f(x) + g(y)y'

PROBLEM STATEMENT

Show that any equation that can be written in the form

$$f(x) + g(y)\frac{dy}{dx} = 0$$

is exact, and find its solution in terms of integrals of f and g. Hence find the solutions of

(i) $V'(x) + 2y\dfrac{dy}{dx} = 0$,
(ii) $\left[\dfrac{1}{y} - a\right]\dfrac{dy}{dx} + \dfrac{2}{x} - b = 0$, for $x, y > 0$.

Source: ProblemsCh2.pdf, Exercise 6 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

General form: $M(x,y) + N(x,y)\,y' = 0$ with $M = f(x)$ depending on x only and $N = g(y)$ depending on y only. Assume f continuous on an interval I_x and g continuous on an interval I_y, so that M and N are continuous on the rectangle $\Omega = I_x \times I_y$ (a simply connected set).

- (i) $f(x) = V'(x)$, with $V \in C^1$ so that V' is continuous; $g(y) = 2y$. Rectangle: I_x any interval, $I_y = \mathbb{R}$.
- (ii) $f(x) = \frac{2}{x} - b$, $g(y) = \frac{1}{y} - a$. Continuous on $x > 0$ and $y > 0$ (the stated domain), so $\Omega = (0,\infty) \times (0,\infty)$.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: the exactness criterion $M_y = N_x$ (Robinson, Section 10.1) is the hypothesis for a potential function F with $F_x = M$, $F_y = N$ to exist on Ω. Here both partial derivatives are trivial, so the criterion can be checked at once, and the potential can be constructed by integrating each variable separately.

Construction: integrate $F_x = f(x)$ in x, then fix the y-dependent integration function $h(y)$ from $F_y = g(y)$. Level sets of F are the solutions.

METHOD OF THE CLASS NOTES [NOTES, PP. 3-4]

In the notation of the notes the equation is $f + g\,\frac{dy}{dx} = 0$ with $f = f(x)$ and $g = g(y)$, so $\frac{\partial f}{\partial y} = 0 = \frac{\partial g}{\partial x}$ and the theorem of the notes (with f, g continuous) gives exactness. The construction of the notes then reads $F = \int f(x)\,dx + \varphi(y)$, $F_y = \varphi'(y) = g(y)$, so $F = \int f(x)\,dx + \int g(y)\,dy = C$: the separable formula of p. 3 written as a level set.

PHASE 3: STEP-BY-STEP DERIVATION

General case

$M = f(x)$ does not depend on y, and $N = g(y)$ does not depend on x:

$$\frac{\partial M}{\partial y} = 0 = \frac{\partial N}{\partial x}.$$

The exactness condition holds, so the equation is exact. Construct F: from $F_x = f(x)$,

$$F(x, y) = \int_{x_0}^{x} f(s)\,ds + h(y).$$

Differentiate in y: $F_y = h'(y)$, and this must equal $N = g(y)$. So $h'(y) = g(y)$ and

$$h(y) = \int_{y_0}^{y} g(r)\,dr.$$

Hence the solutions are the level curves

$$\boxed{\int_{x_0}^{x} f(s)\,ds + \int_{y_0}^{y} g(r)\,dr = C}, \qquad C \in \mathbb{R}.$$

If a solution passes through (x_0, y_0), then $C = 0$ with these lower limits (Barrow's rule fixes the constant). The same result follows from separation of variables: $g(y)\,dy = -f(x)\,dx$, then integrate both sides.

Part (i): $V'(x) + 2y\,y' = 0$

Here $f = V'$ and $g = 2y$. The integrals are $\int V'(x)\,dx = V(x)$ and $\int 2y\,dy = y^2$. Therefore

$$V(x) + y^2 = C.$$

Equivalent explicit form: $y = \pm\sqrt{C - V(x)}$, defined where $V(x) \le C$. The equation says $\frac{d}{dx}\left[V(x) + y^2\right] = V' + 2yy' = 0$, i.e. $V + y^2$ is conserved along the solution.

Part (ii): $\left[\frac{1}{y} - a\right] y' + \frac{2}{x} - b = 0, \;\; x, y > 0$

Here $f(x) = \frac{2}{x} - b$ and $g(y) = \frac{1}{y} - a$. Integrate each, using $x > 0$ and $y > 0$ so no absolute values are needed:

$$\int \left(\frac{2}{x} - b\right) dx = 2\ln x - bx, \qquad \int \left(\frac{1}{y} - a\right) dy = \ln y - ay.$$

The solutions are therefore

$$2\ln x - bx + \ln y - ay = C.$$

Combine the logarithms and exponentiate: $\ln(x^2 y) = C + bx + ay$, so with $K = e^C > 0$,

$$x^2\, y\, e^{-bx - ay} = K.$$

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check by implicit differentiation:

- General case: $\frac{d}{dx}\left[\int_{x_0}^{x} f\,ds + \int_{y_0}^{y(x)} g\,dr\right] = f(x) + g(y)\,y'$, using the Fundamental Theorem of Calculus and the chain rule; this vanishes by the ODE.
- (i) $\frac{d}{dx}\left[V(x) + y^2\right] = V'(x) + 2y\,y' = 0$.
- (ii) $\frac{d}{dx}\left[2\ln x - bx + \ln y - ay\right] = \frac{2}{x} - b + \left(\frac{1}{y} - a\right)y' = 0$, the original equation.

Dimensional remark for (ii): $\ln x$ and $\ln y$ require dimensionless arguments, so x and y are understood as measured in fixed reference units; then bx and ay are dimensionless, which means $[b] = [x]^{-1}$ and $[a] = [y]^{-1}$, consistent with the terms $\frac{2}{x} - b$ and $\frac{1}{y} - a$ being summed.

Domain: in (ii) the level set $x^2 y e^{-bx-ay} = K$ in the open quadrant is a union of curves; y is a function of x locally wherever $F_y = \frac{1}{y} - a \neq 0$, i.e. away from $y = 1/a$ when $a > 0$.

RESULTS

General form: $\int f\,dx + \int g\,dy = C$. (i) $V(x) + y^2 = C$. (ii) $2\ln x - bx + \ln y - ay = C$, i.e. $x^2 y\,e^{-bx-ay} = K > 0$.